

Technical University of Denmark
Department of Applied Mathematics and Computer Science

02170 Database Systems

Practice Examination Part 1 — Exam B

Aids allowed: All
Duration: 3 hours
Weighting: Part 1: 50%, Part 2: 50%

Note: unless explicitly asked for, SQL queries should not return tables having duplicate rows.

A database for a Hospital Network

A database named **HospitalDB** contains information about hospitals, doctors, studies, and research projects.

A **Hospital** has a unique name (*hname*) and is located in a country (*country*). A **Doctor** has a unique id (*did*), a name (*dname*), and works at a single hospital (identified by *hname*). A **Study** has a unique id (*sid*), a title (*stitle*), and a publication year (*syear*).

The **conducts** relation records which doctors have conducted which studies. The **ResearchGroup** relation represents research groups, each hosted at a hospital. The **member_of** relation records which doctors are members of which research groups.

Relation schemas:

Hospital(hname, country)

Study(sid, stitle, syear)

ResearchGroup(gname, hname)

Doctor(did, dname, hname)

conducts(did, sid)

member_of(gname, did)

Hospital

hname	country
Rigshospitalet	Denmark
OUH	Denmark
Karolinska	Sweden
Oslo Univ	Norway

Doctor

did	dname	hname
10	Mette	Rigshospitalet
20	Lars	Rigshospitalet
30	Sofia	Karolinska
40	Jonas	OUH
50	Nina	Oslo Univ

Study

sid	stitle	syear
1	Cancer Research	2023
2	Heart Study	2022
3	Brain Connectivity	2024

conducts

did	sid
10	1
20	1
20	2
30	2
40	3

ResearchGroup

gname	hname
CardioGroup	Rigshospitalet
NeuroGroup	Karolinska
OncologyTeam	OUH

member_of

gname	did
CardioGroup	10
CardioGroup	20
NeuroGroup	30
OncologyTeam	40

Questions

Question 1

Write an SQL query that returns a table containing, for each doctor who works at a **Danish** hospital and who has **never conducted any study**, his/her **dname** and **hname**.

Question 2

Write an SQL query that returns a table containing, for each study conducted by **2 or more doctors**, the **sid**, **stitle**, **syear**, and the total number of doctors who conducted it. Name the count **numDoctors**.

Question 3

Write an SQL query that returns a table containing, for each research group, the **gname** and the number of **distinct studies conducted by at least one member of that group**. Name the count **numStudies**.

Important: every research group must appear in the result, even those whose members have not conducted any study.

Question 4

Create an SQL function named **studyCount**, which takes the **did** (of type **INT**) of a doctor as argument and returns the number of studies conducted by that doctor. If the doctor

does not exist, it should return 0. The returned type should be `INT`.

Do not include any `DROP FUNCTION` or `SELECT` statements using the function.

Answers Template (answers_B.sql)

```
-- Student ID:  
USE HospitalDB;  
  
-- q1:  
-- q2:  
-- q3:  
-- q4:
```